**Proof.** If $V \in D$, the number $\frac{1}{p}\dim(V)\chi_V(g)$ is an algebraic integer, so

$$a = \sum_{V \in D} \frac{1}{p}\dim(V)\chi_V(g)$$

is an algebraic integer.

Now, by (5.4.1), we have

$$\begin{aligned} 0 &= \chi_{\mathbb{C}}(g) + \sum_{V \in D} \dim V \chi_V(g) + \sum_{V \in N} \dim V \chi_V(g) \\ &= 1 + pa + \sum_{V \in N} \dim V \chi_V(g). \end{aligned}$$

This means that the last summand is nonzero. □

Now pick $V \in N$ such that $\chi_V(g) \neq 0$; it exists by Lemma 5.4.7. Theorem 5.4.4 implies that $g$ (and hence any element of $C$) acts by a scalar in $V$. Now let $H$ be the subgroup of $G$ generated by elements $ab^{-1}$, $a, b \in C$. It is normal and acts trivially in $V$, so $H \neq G$, as $V$ is nontrivial. Also $H \neq 1$, since $|C| > 1$. □

**Proof of Burnside's theorem.** Assume Burnside's theorem is false. Then there exists a nonsolvable group $G$ of order $p^a q^b$. Let $G$ be the smallest such group. Then $G$ is simple, and by Theorem 5.4.6, it cannot have a conjugacy class of order $p^k$ or $q^k$, $k \geq 1$. So the order of any conjugacy class in $G$ either equals 1 or is divisible by $pq$. Adding the orders of conjugacy classes and equating the sum to $p^a q^b$, we see that there has to be more than one conjugacy class consisting just of one element. So $G$ has a nontrivial center, which gives a contradiction. □

## 5.5. Historical interlude: William Burnside and intellectual harmony in mathematics

While at Cambridge, William Burnside (1852–1927) distinguished himself in rowing; his other achievements include emerging from the 1875 Mathematical Tripos as Second Wrangler and then beating First Wrangler in an even more grueling mathematical competition for the Smith Prize. Afterwards, he taught at Cambridge as a mathematics lecturer and a coach for both the Math Tripos and for the rowing